

Program: Diploma in Microelectronics	
Course Code: 5493A	Course Title: Low Power VLSI
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To know about various short channel effects in MOSFET
- To know about various sources of power dissipation in MOSFET
- To understand various power reduction techniques in circuits.
- To implement logic using various non clocked and clocked design styles.

Course Prerequisites:

Topic	Course code	Course Title	Semester
MOS physics, Static and Dynamic CMOS logic design		VLSI Design	4
		Digital System Design	5

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Identify various short channel effects in MOSFET	15	Understanding
CO2	Understand various sources of power dissipation in MOSFET	14	Understanding
CO3	Understand various power reduction techniques in circuits	14	Understanding
CO4	Apply various non clocked and clocked design styles for logic implementation	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify various short channel effects in MOSFET		
M1.01	Explain the need for low power circuit design	2	Understanding
M1.02	Explain the MIS structure	3	Understanding
M1.03	Discuss short channel effects in MOSFET and methods to minimize SCE	8	Understanding
M1.04	Describe the design issues with deep submicron transistor and leakage mechanisms	2	Understanding
Contents: Physics of Power dissipation in MOSFET devices Need for low power circuit design, MIS Structure, Short channel effects (SCE) - Channel Length Modulation, Drain Induced Barrier Lowering, punch through, velocity saturation, surface scattering, Impact ionization, Hot electron effects, Minimizing SCE, Deep submicron transistor design issues, Leakage mechanisms in short channel transistors.			
CO2	Understand various sources of power dissipation in MOSFET		
M2.01	Discuss the sources of static power dissipation	3	Understanding
M2.02	Explain switching power dissipation	4	Understanding
M2.03	Describe short circuit power dissipation	4	Understanding
M2.04	Discuss glitching power dissipation	3	Understanding
	Series Test - I	1	

Contents:**Sources of power dissipation in CMOS (No derivation required)**

Static Power Dissipation - Leakage Power Dissipation

Dynamic Power Dissipation: Switching power dissipation, Short Circuit power dissipation: Short Circuit Current of Inverter, Short circuit current dependency with input and output load, Glitching Power dissipation

CO3	Understand various power reduction techniques in circuits		
M3.01	Explain the supply voltage scaling approaches for power reduction	2	Understanding
M3.02	Discuss channel engineering for leakage power reduction	2	Understanding
M3.03	Explain the circuit techniques for leakage power reduction	6	Understanding
M3.04	Describe the techniques for dynamic power reduction	4	Understanding

Contents:**Power Reduction Techniques**

Supply voltage Scaling Approaches: Multi VDD and Dynamic VDD

Leakage power reduction Techniques : Channel Engineering - Retrograde Doping, Halo Doping, Circuit Techniques - Transistor stacking, VTCMOS, MTCMOS, DTCMOS, Power gating

Dynamic power reduction Techniques : Clock gating, Gate sizing, Logic restructuring, Pin swapping

CO4	Apply various non-clocked and clocked design styles for logic implementation		
M4.01	Apply fully complementary logic for implementing Boolean expressions	7	Applying
M4.02	Apply NMOS logic and pseudo-NMOS logic for function implementation	4	Applying
M4.03	Apply differential cascade voltage switch logic for implementing NAND/AND and NOR/OR circuits	4	Applying
	Series Test - I	1	

Contents:

Circuit design style - Non clocked circuit design style - fully complementary logic,

NMOS and pseudo–NMOS logic, differential cascade voltage switch logic (DCVS) - DCVS NAND/AND, DCVS NOR/OR circuits.

Text /Reference:

T/R	Book Title/Author
T1	Kaushik Roy, Sharat C Prasad, Low power CMOS VLSI circuit design, Wiley India, 2000
T2	John P. Uyemura, CMOS Logic Circuit Design, Kluwer Academic Publishers, Newyork, 2001
T3	Ajit Pal, Low-Power VLSI Circuits and Systems, Springer, 2015, DOI : 10.1007/978-81-322-1937-8
R1	Abdellatif Bellaouar, Mohamed I Elmasry, Low power digital VLSI design, Kluwer Academic, 1995
R2	Kaushik Roy, Hamid Mahmoodi-Meimand and Saibal Mukhopadhyay, "Leakage control for deep-submicron circuits", Proc. SPIE 5117, VLSI Circuits and Systems, (21 April 2003); https://doi.org/10.1117/12.498181

Online Resources:

Sl. No.	Website Link
1	https://www.academia.edu/77767705/Leakage_control_for_deep_submicron_circuits
2	https://www.scribd.com/document/61476406/paper-C7
3	https://www.cerc.utexas.edu/~jaa/vlsi/lectures/3-2.pdf
4	https://www.cerc.utexas.edu/~jaa/vlsi/lectures/12-1.pdf
5	https://vlsi-iitg.vlabs.ac.in/LogicGates_theory.html